

Area Methods in Computational Geometry

1 Overview

Area Methods

Area methods use coordinate formulas to solve two common computational-geometry tasks:

- compute the area of a polygon;
- decide whether a query point is inside, outside, or on the boundary of a polygon.

Method	Input	Output
Triangle area	Three points A, B, C	Area of $\triangle ABC$.
Shoelace formula	Polygon vertices in boundary order	Polygon area.
Area-sum method	Polygon and query point Q	Inside, outside, or boundary decision.

Recommended Order

For point-in-polygon by area, first compute the polygon area using Shoelace. Then compute the sum of triangle areas formed by the query point and each polygon edge.

2 Base Formula: Triangle Area

For three points

$$A = (x_1, y_1), \quad B = (x_2, y_2), \quad C = (x_3, y_3),$$

the triangle area is:

Triangle Area

$$\text{Area}(\triangle ABC) = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$$

Equivalently, using orientation:

$$\text{Area}(\triangle ABC) = \frac{1}{2} |\text{orient}(A, B, C)|.$$

Example

Problem

Find the area of

$$A = (1, 1), \quad B = (4, 1), \quad C = (2, 3).$$

$$\overrightarrow{AB} = (4 - 1, 1 - 1) = (3, 0), \quad \overrightarrow{AC} = (2 - 1, 3 - 1) = (1, 2).$$

$$\text{orient}(A, B, C) = 3(2) - 0(1) = 6.$$

$$\text{Area}(\triangle ABC) = \frac{1}{2}|6| = 3.$$

Answer

$$\text{Area}(\triangle ABC) = 3 \text{ square units.}$$

3 Method 1: Shoelace Formula

Use Shoelace when the polygon vertices are known in clockwise or counterclockwise boundary order:

$$P_0 = (x_0, y_0), \quad P_1 = (x_1, y_1), \quad \dots, \quad P_{n-1} = (x_{n-1}, y_{n-1}).$$

Use wraparound:

$$P_n = P_0.$$

Shoelace Formula

$$A = \frac{1}{2} \left| \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) \right|.$$

Vertex Order

The vertices must be listed around the boundary. A random order of the same points does not give the correct polygon area.

4 Example 1: Shoelace Formula

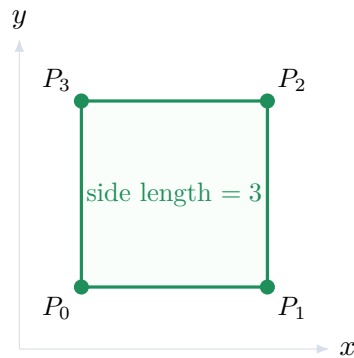
Problem

Find the area of the square

$$P_0 = (1, 1), \quad P_1 = (4, 1), \quad P_2 = (4, 4), \quad P_3 = (1, 4).$$

Solution

The vertices are already listed in counterclockwise boundary order. Draw the square:



Use

$$A = \frac{1}{2} \left| \sum_{i=0}^3 (x_i y_{i+1} - x_{i+1} y_i) \right|, \quad P_4 = P_0.$$

Edge	Shoelace term	Value
$P_0 P_1$	$x_0 y_1 - x_1 y_0 = 1(1) - 4(1)$	-3
$P_1 P_2$	$x_1 y_2 - x_2 y_1 = 4(4) - 4(1)$	12
$P_2 P_3$	$x_2 y_3 - x_3 y_2 = 4(4) - 1(4)$	12
$P_3 P_0$	$x_3 y_0 - x_0 y_3 = 1(1) - 1(4)$	-3

So the signed double-area is

$$-3 + 12 + 12 - 3 = 18.$$

Therefore:

$$A = \frac{1}{2} |18| = 9.$$

Answer

Area = 9 square units.

5 Example 2: Shoelace Table Form

The same calculation can be done using the table form. Write the first point again at the bottom:

Point	x_i	y_i
P_0	1	1
P_1	4	1
P_2	4	4
P_3	1	4
P_0	1	1

Add the downward diagonal products:

$$1 \cdot 1 + 4 \cdot 4 + 4 \cdot 4 + 1 \cdot 1 = 1 + 16 + 16 + 1 = 34.$$

Add the upward diagonal products:

$$1 \cdot 4 + 1 \cdot 4 + 4 \cdot 1 + 4 \cdot 1 = 4 + 4 + 4 + 4 = 16.$$

Then:

$$A = \frac{1}{2}|34 - 16| = \frac{1}{2}(18) = 9.$$

Table Form Result

The table form gives the same area: 9.

6 Method 2: Area-Sum Point Test

Let the polygon be

$$P_0P_1 \cdots P_{n-1},$$

and let Q be the query point. Connect Q to every polygon edge:

$$\triangle QP_0P_1, \quad \triangle QP_1P_2, \quad \dots, \quad \triangle QP_{n-1}P_0.$$

Compute

$$S = \sum_{i=0}^{n-1} \text{Area}(\triangle QP_iP_{i+1}), \quad P_n = P_0.$$

Let A be the polygon area. Then:

Area-Sum Decision Rule

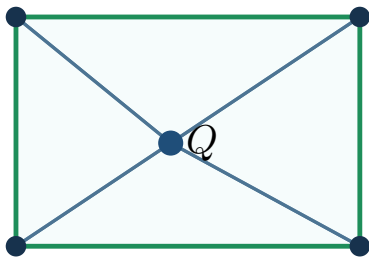
$$\begin{aligned} S = A &\implies Q \text{ is inside or on the boundary,} \\ S > A &\implies Q \text{ is outside.} \end{aligned}$$

If $S = A$ and one of the triangle areas is 0, then Q lies on a boundary edge. If $S = A$ and no triangle area is 0, then Q is strictly inside.

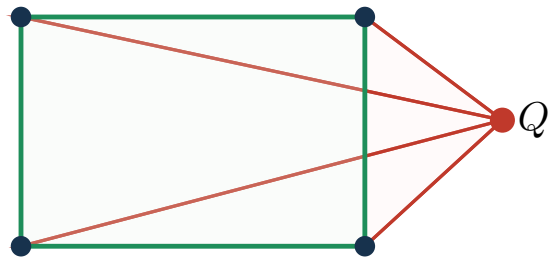
7 Why Area-Sum Works

If Q is inside the polygon, the triangles from Q to each edge exactly partition the polygon. Their areas add up to the polygon area.

If Q is outside, the triangles overlap or include extra region outside the polygon. Then the area sum becomes larger than the polygon area.



inside: triangles fill polygon



outside: extra area is counted

Numerical Tolerance

In numerical programs, avoid checking $S = A$ using exact floating-point equality. Use

$$|S - A| \leq \varepsilon.$$

8 Example 3: Inside Point

Problem

Use the same square and test

$$Q = (2, 2).$$

Solution

The polygon area from the shoelace formula is

$$A = 9.$$

Now split the polygon into four triangles:

$$\triangle QP_0P_1, \quad \triangle QP_1P_2, \quad \triangle QP_2P_3, \quad \triangle QP_3P_0.$$

Compute each triangle area using

$$\text{Area}(\triangle ABC) = \frac{1}{2} |\text{orient}(A, B, C)|.$$

Triangle	Orientation calculation	Area
$\triangle QP_0P_1$	$(1 - 2)(1 - 2) - (1 - 2)(4 - 2) = (-1)(-1) - (-1)(2) = 3$	$\frac{3}{2}$
$\triangle QP_1P_2$	$(4 - 2)(4 - 2) - (1 - 2)(4 - 2) = 2(2) - (-1)(2) = 6$	3
$\triangle QP_2P_3$	$(4 - 2)(4 - 2) - (4 - 2)(1 - 2) = 2(2) - 2(-1) = 6$	3
$\triangle QP_3P_0$	$(1 - 2)(1 - 2) - (4 - 2)(1 - 2) = (-1)(-1) - 2(-1) = 3$	$\frac{3}{2}$

Therefore:

$$S = \frac{3}{2} + 3 + 3 + \frac{3}{2} = 9.$$

Since

$$S = A = 9,$$

the point is inside or on the boundary. Since no triangle area is zero, Q is strictly inside.

Answer

$Q = (2, 2)$ is inside the square.

9 Example 4: Outside Point

Problem

Use the same square and test

$$R = (5, 2).$$

Solution

Again, the polygon area is

$$A = 9.$$

Compute the four triangle areas:

$$\text{Area}(\triangle RP_0P_1) = \frac{1}{2}|(-4)(-1) - (-1)(-1)| = \frac{3}{2},$$

$$\text{Area}(\triangle RP_1P_2) = \frac{1}{2}|(-1)(2) - (-1)(-1)| = \frac{3}{2},$$

$$\text{Area}(\triangle RP_2P_3) = \frac{1}{2}|(-1)(2) - (2)(-4)| = 3,$$

$$\text{Area}(\triangle RP_3P_0) = \frac{1}{2}|(-4)(-1) - (2)(-4)| = 6.$$

So:

$$S = \frac{3}{2} + \frac{3}{2} + 3 + 6 = 12.$$

Since

$$S = 12 > A = 9,$$

the triangle areas count extra region outside the square.

Answer

$R = (5, 2)$ is outside the square.

10 Example 5: Boundary Point

Problem

Use the same square and test

$$T = (3, 1).$$

Solution

The point T lies on the bottom edge P_0P_1 . The triangle using that edge has zero area:

$$\text{Area}(\triangle TP_0P_1) = \frac{1}{2}|\text{orient}(T, P_0, P_1)| = 0.$$

The remaining triangle areas still fill the square:

$$\text{Area}(\triangle TP_1P_2) = \frac{3}{2},$$

$$\text{Area}(\triangle TP_2P_3) = \frac{9}{2},$$

$$\text{Area}(\triangle TP_3P_0) = 3.$$

Therefore:

$$S = 0 + \frac{3}{2} + \frac{9}{2} + 3 = 9 = A.$$

Answer

$T = (3, 1)$ is on the boundary of the square.

11 Complexity

Both methods scan polygon edges once.

Method	Time complexity
Area-sum inclusion test	$\mathcal{O}(n)$ for n polygon edges.
Shoelace area formula	$\mathcal{O}(n)$ for n polygon vertices.

12 Final Review

Area Methods Summary

$$S = \sum \text{Area}(\triangle QP_iP_{i+1}).$$

$$S = A \Rightarrow Q \text{ is inside or on boundary}; \quad S > A \Rightarrow Q \text{ is outside.}$$

$$A = \frac{1}{2} \left| \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) \right|.$$

Use $P_n = P_0$ for wraparound.